

The agent will Plan → Build → Test → Review → Audit in a loop across the system, providing you with a screenshot, a checklist, and the URL to open.

STEP 1

Drop These 5 Files Into Your Project

Copy to the exact paths shown — this is the entire setup

Do this once. After you drop these files, you never touch them again. The agent reads them automatically and uses them to run the entire build.

File to create	Where it goes	What it does
GEMINI.md	surgimart/GEMINI.md	Antigravity reads this automatically. Contains the loop instructions, all coding rules, and the notification format.
tasks.md	surgimart/tasks.md	37 tasks across 10 phases. The agent picks the first TODO, does the full cycle, marks DONE, moves on.
.agents/workflows/surgimart-build.md	surgimart/.agents/workflows/surgimart-build.md	The /surgimart-build workflow. You type this once in Agent Manager to start the loop.
.agents/skills/surgimart-loop/SKILL.md	surgimart/.agents/skills/surgimart-loop/SKILL.md	Contains HANDOFF.md and MISSING.md templates. Agent writes these when done.
.agents/skills/surgimart-testing/SKILL.md	surgimart/.agents/skills/surgimart-testing/SKILL.md	All test patterns, commands, MSW setup, coverage rules. Agent loads when writing tests.
.agents/skills/surgimart-audit/SKILL.md	surgimart/.agents/skills/surgimart-audit/SKILL.md	Complete audit checklist, all 15 security findings, fix patterns. Agent loads each audit step.
agent-log.md	surgimart/agent-log.md	Empty file. Agent appends progress after every completed task. You can watch it in real-time.

Quick file structure check

Folder structure to create

surgimart/

■■■ GEMINI.md ← Antigravity reads this automatically

■■■ tasks.md ← 37 tasks, agent updates status as it works

■■■ agent-log.md ← progress log (append-only)

■■■ .agents/

■■■ workflows/

■ ■■■ surgimart-build.md ← trigger with /surgimart-build

■■■ skills/

■■■ surgimart-loop/SKILL.md ← HANDOFF.md + MISSING.md templates

■■■ surgimart-testing/SKILL.md ← all test patterns + commands

■■■ surgimart-audit/SKILL.md ← security checklist + fix patterns

STEP 2

Start the Autonomous Loop

One command. Then watch the Agent Manager dashboard.

#	Action	Detail
1	Open Antigravity	Launch the app. Go to Agent Manager (the default view).
2	Open your workspace	Click Workspaces → select your surgimart/ project folder.
3	Set the model	Click the Model dropdown → select Gemini 3 Pro or Claude Sonnet 4.5.
4	Set mode to Planning	Click the mode dropdown → select Planning (not Fast). Planning mode writes implementation plans as Artifacts before touching code.
5	Set terminal policy	Customizations → Terminal Execution → Always Proceed. This lets the agent run npm test, npm build, etc without asking permission.
6	Set review policy	Customizations → Review Policy → Agent Decides. Agent will only stop for review when it genuinely needs a decision.
7	Type the command	In the chat input type exactly: /surgimart-build Press Enter. That's it.
8	Walk away	The agent will now run all 37 tasks autonomously. Come back when it notifies you.

Do not send any other messages while the loop is running. The agent is working. You can watch the Artifacts panel on the right to see implementation plans and screenshots. The terminal panel shows all commands being run. If you need to stop: click the Stop button in Agent Manager. To resume: type /surgimart-build again — it reads tasks.md and skips DONE tasks.

STEP 3

How the Loop Works

What the agent does for every single task — automatically

Phase	What happens	Output (Artifact)
1 — PLAN	Agent reads tasks.md, picks next TODO task. Reads the relevant skill. Writes an implementation plan: what files to create, what the approach is, what the tests will verify. Never writes a single line of code yet.	Artifact: "Plan: T-XX Task Name" Lists files, approach, test strategy, risk level.
2 — BUILD	Agent writes the code following all rules in GEMINI.md. Every function is complete — no stubs, no placeholders, no TODOs. Every new file gets a corresponding test file.	Code diffs visible in the Editor panel.
3 — TEST	Agent runs 5 checks in sequence: 1. npx tsc --noEmit (types) 2. npm run lint (style) 3. npm test (unit + component) 4. npm test --coverage (≥80% required) 5. npm run build (build check) If any fails: fix it. Re-run. Up to 5 attempts.	Terminal output visible. All must show green before moving on.
4 — REVIEW	Agent starts the dev server. Opens http://localhost:3000 in the integrated Chrome browser. Checks the UI looks correct. Opens DevTools and verifies zero console errors. Takes a screenshot.	Artifact: "Review: T-XX" — screenshot of the feature working.
5 — AUDIT	Agent runs npm audit (zero high vulns), greps for "any" (zero allowed), greps for console.log (zero allowed), greps for @ts-ignore (zero allowed).	Artifact: "Audit: T-XX" — clean report.
6 — LOG	Agent marks task DONE in tasks.md. Appends to agent-log.md with timestamp, files changed, test results, coverage, audit status.	tasks.md updated. agent-log.md entry added.
7 — REPEAT	Immediately picks next TODO task from tasks.md. Runs the full loop again. No pause, no asking. All 37 tasks this way.	37 implementation plans + 37 review screenshots.
8 — NOTIFY	After T-37: runs make test-ci (full suite). Takes homepage screenshot. Writes HANDOFF.md and MISSING.md. Posts completion notification to you.	Final Artifact: full test report + homepage screenshot + notification.

STEP 4

The 37 Tasks — What Gets Built

10 phases in order of dependency

Task	Name	Phase	Tests
T-01	Fix jest.config.ts typo (broken test suite)	1 — Foundation	npm test works
T-02	Fix docker-compose bugs (version key + Redis URL)	1 — Foundation	docker compose config clean
T-03	Homepage calls getFeaturedProducts (BUG-1)	1 — Foundation	ProductGrid renders
T-04	Navbar auth-aware (BUG-2)	1 — Foundation	Shows avatar when logged in
T-05	ProductGrid static Tailwind class (BUG-4)	1 — Foundation	Grid renders in prod build
T-06	All TypeScript types in src/types/index.ts	1 — Foundation	tsc --noEmit clean
T-07	All lib singletons (prisma, redis, stripe, etc.)	1 — Foundation	Imports work, server-only
T-08	CSRF + rate limiting lib	1 — Foundation	403 on CSRF, 429 on rate limit
T-09	Hooks (useDebounce, useHasMounted, useScroll)	1 — Foundation	Hook tests pass
T-10	5 Zustand stores	1 — Foundation	All store actions verified
T-11	Fix payment intent (never trust client amount)	2 — Security	C-1, C-5 audit fixed
T-12	Fix Stripe webhook (sync, null check, updateMany)	2 — Security	C-2, C-3, C-6 fixed
T-13	Fix order number (crypto.randomUUID)	2 — Security	10K nums, zero duplicates
T-14	Apply CSRF + rate limits to all routes	2 — Security	All routes protected
T-15	Split Sanity clients (read/write)	2 — Security	Token not in client bundle
T-16	UI: Skeleton, StarRating, CookieBanner	3 — Components	Component tests pass
T-17	Layout: AnnouncementBar, Footer	3 — Components	Renders, all links correct

Task	Name	Phase	Tests
T-18	ProductCard (memo, convertedPrice prop)	3 — Components	Add to cart, wishlist, compare
T-19	ProductGrid + ZoomModal + QuoteModal + ComparisonBar	3 — Components	All interactions work
T-20	CartDrawer (slide-in, qty, shipping bar)	3 — Components	E2E cart flow passes
T-21	SearchAutocomplete (Algolia, debounced)	3 — Components	Dropdown shows results
T-22	6 homepage section components	3 — Components	All sections render
T-23	Checkout flow (3 steps + Stripe)	3 — Components	Full checkout E2E pass
T-24	All 7 API routes	4 — API Routes	Unit + integration tests
T-25	All 13 pages	5 — Pages	All routes return 200, no errors
T-26	Error + loading + not-found pages	5 — Pages	Custom pages render
T-27	Complete Prisma schema	6 — Database	prisma validate + generate
T-28	Database seed file	6 — Database	make db-seed, test data created
T-29	Sanity schemas + studio config	6 — Database	npx sanity check clean
T-30	Algolia sync scripts	6 — Database	--dry-run exits 0
T-31	5 React Email templates	7 — Emails	make email-preview, all render
T-32	Full Playwright E2E suite	8 — E2E Tests	All scenarios pass
T-33	Accessibility audit (jest-axe)	9 — Production	Zero axe violations
T-34	SEO metadata pass (all pages)	9 — Production	npm run build clean
T-35	Live currency exchange rates	9 — Production	Live + fallback both work
T-36	Docker validation	9 — Production	make docker-health green
T-37	Final verification + notify user	10 — Final	All green + HANDOFF.md written

What Happens When Done

The notification you will receive in Agent Manager

[illegible]

File	Contains
HANDOFF.md	Complete manual testing checklist. Test credentials (admin@surgimart.com / Admin123!, buyer@test.com / Test123!). Test order SM-TEST0001. Stripe test card. Step-by-step instructions for every feature. App URLs. Service URLs.
MISSING.md	Everything that requires your API keys and cannot be set up autonomously. Each item has: what it is, where to get the key, exact env var name, and what works without it (demo mode). See next page for the full list.

STEP 6

What You Must Add Manually

API keys that require your accounts — the agent cannot create these

The agent builds everything it can without your accounts. The app runs in demo mode without keys. For full functionality you need these. The MISSING.md file in your project explains each one step by step.

#	Service	Required For	Get It At	Env Var(s)	Without It
1	Stripe	Real payments	dashboard.stripe.com	STRIPE_SECRET_KEY NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY STRIPE_WEBHOOK_SECRET	Payments broken
2	Sanity.io	Product catalog (adding real products)	sanity.io → New Project	NEXT_PUBLIC_SANITY_PROJECT_ID NEXT_PUBLIC_SANITY_DATASET SANITY_API_TOKEN SANITY_WEBHOOK_SECRET	No products (uses seed data only)
3	Google OAuth	Google login button	console.cloud.google.com → Credentials	GOOGLE_CLIENT_ID GOOGLE_CLIENT_SECRET	Only email/password login works
4	NextAuth Secret	Session encryption	openssl rand -base64 32	NEXTAUTH_SECRET	Auth broken completely
5	Resend	Confirmation emails	resend.com → API Keys	RESEND_API_KEY RESEND_FROM (verified domain)	No emails sent (orders still work)
6	Algolia	Search autocomplete	algolia.com → Create App	NEXT_PUBLIC_ALGOLIA_APP_ID NEXT_PUBLIC_ALGOLIA_SEARCH_KEY ALGOLIA_ADMIN_KEY	Search bar empty (no autocomplete)
7	Open Exchange Rates	Live currency conversion	openexchangerates.org	EXCHANGE_RATES_APP_ID	Hardcoded rates used (still works)
8	DB + Redis passwords	Docker services start	Generate locally	DB_PASSWORD REDIS_PASSWORD	Docker services fail to start

Step-by-step: add keys after the agent finishes

After the agent notifies you: add keys and run these

1. The agent creates .env.example — copy it

```
cp .env.example .env.local
```

2. Open .env.local and fill in your keys

```
nano .env.local # or open in Antigravity editor
```



```
# 3. Restart to pick up new env vars

make docker-dev

# 4. Add products to Sanity Studio

open http://localhost:3333

# Sign in → Products → New Product → add at least 5

# 5. Push products to Algolia search

make algolia-sync

# 6. Set up Stripe webhook for local testing

stripe login

stripe listen --forward-to localhost:3000/api/stripe/webhook

# Copy whsec_... to STRIPE_WEBHOOK_SECRET in .env.local

# Restart: make docker-dev
```

STEP 7

Watching Progress While It Runs

How to monitor the agent without interfering

What to watch	Where in Antigravity	What you see
Artifacts panel	Right side of Agent Manager	Every implementation plan, every browser screenshot, every test result. A new artifact appears for each phase of each task.
Terminal panel	Bottom of the Editor view	Live output of every npm test, npm build, tsc --noEmit. Green output = passing. Red = agent is fixing it.
tasks.md file	Editor view → tasks.md	Watch Status change from TODO → IN_PROGRESS → DONE in real time.
agent-log.md file	Editor view → agent-log.md	Append-only log. Every completed task gets an entry with timestamp, file list, test result, coverage %, audit result.
Agent Manager inbox	Main Agent Manager view	The agent posts status updates here. The final notification appears here too.

If something looks wrong

Situation	What to do
Agent is stuck on a task for a long time	Check the terminal output. If it is in a fix loop, read the error. You can leave feedback on the Artifact — the agent reads it without stopping.
Agent asks you a question	This should not happen (GEMINI.md tells it not to ask). If it does, answer briefly and add "continue autonomously".
You need to stop the agent	Click Stop in Agent Manager. To resume: type /surgimart-build again. It reads tasks.md, skips DONE tasks, continues from where it stopped.
A task is marked BLOCKED	Agent could not solve it after 5 attempts. Read tasks.md — it will have the error explanation. You can fix it manually then retype /surgimart-build.
You want to skip a task	Open tasks.md in the editor and change that task Status to DONE. The agent will skip it on the next run.


```
# STRIPE_SECRET_KEY, NEXT_PUBLIC_STRIPE_PUBLISHABLE_KEY, STRIPE_WEBHOOK_SECRET

# NEXT_PUBLIC_SANITY_PROJECT_ID, SANITY_API_TOKEN, SANITY_WEBHOOK_SECRET

# GOOGLE_CLIENT_ID, GOOGLE_CLIENT_SECRET

# NEXTAUTH_SECRET (openssl rand -base64 32)

# DB_PASSWORD, REDIS_PASSWORD (any strong string)

# RESEND_API_KEY (optional – emails)

# NEXT_PUBLIC_ALGOLIA_APP_ID, ALGOLIA_ADMIN_KEY (optional – search)

# 2. make docker-dev

# 3. open http://localhost:3333 → add products in Sanity Studio

# 4. make algolia-sync

# 5. Follow HANDOFF.md checklist
```